Embeddings and neighbourhoods

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable embedding and neighbourhood record.

Look up vectors,calculate distances,retain ties and explain averaging/fitting limits.

Keep one habit: W1–W3/W5 → selected new candidate table → project embedding ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Continue the fictional robot vocabulary and add WAIT as ID 5. Use the table below, chosen by the lesson author for arithmetic practice. These coordinates are not measured robot properties and were not learned from data.

  Each token ID selects one row; the resulting two-component vector is its embedding in this example.

- The table shape is 6 by 2: six vocabulary entries and **embedding dimension** 2. The ID sequence [2, 4, 0, 0] becomes [[2, 0], [0, 2], [0, 0], [0, 0]], with shape 4 by 2. Its mask remains [1, 1, 0, 0].

  The vector dimension is not the sequence length or vocabulary size.

- A zero PAD vector does not remove the need for masking. Averaging all four vectors above gives [0.5, 0.5]; averaging only the two real vectors gives [1, 1].

  UNK is a real position when it replaces an unknown symbol, so its vector is not automatically excluded.

Sequence representations

My remembered idea, example and correction

Training a small image model

My remembered idea, example and correction

Image tensors and channels

My remembered idea, example and correction

Lesson reference for workbook practice

Symbol	ID	Vector
PAD	0	[0, 0]
UNK	1	[-1, -1]
GO	2	[2, 0]
TURN	3	[1, 1]
STOP	4	[0, 2]
WAIT	5	[0, 1]

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Original mapping:PAD=0→[0, 0], UNK=1→[-1, -1], GO=2→[2, 0], TURN=3→[1, 1], STOP=4→[0, 2], WAIT=5→[0, 1]. The numeric table has shape 6×2. GO STOP with two padding slots looks up[[2, 0], [0, 2], [0, 0], [0, 0]], mask[1, 1, 0, 0], shape 4×2. Its masked vector mean is[1, 1]; the all-slot mean is[.5, .5]. A real UNK position stays included. Squared distance sums matching-coordinate squared differences.

  For query GO[2, 0], exclude itself and reserved entries:TURN distance 2, STOP distance 8, WAIT distance 5, so TURN is nearest. For arbitrary query[1, 0], all four regular candidates are eligible:distances GO=1, TURN=1, STOP=5, WAIT=2; return both GO and TURN. For query STOP, distances to GO/TURN/WAIT are 8/2/1. The original coordinates were hand-chosen.

  The isolated GO-row objective toward[0, 2]has gradient[4, -4]; rate.25 gives[1, 1]and loss 8→2. Unchanged TURN also has[1, 1], so the updated rows collide without establishing useful meaning. GO STOP and STOP GO have the same mean[1, 1], also equal to the single TURN vector. Save/change the vocabulary mapping and table rows consistently. The pictured P/Q/R table uses separately supplied coordinates.

- Using the original table, embed TURN WAIT STOP with one right-padding slot. Give the vector sequence, mask and shape.

> **My reasoning and evidence**

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Original mapping:PAD=0→[0, 0], UNK=1→[-1, -1], GO=2→[2, 0], TURN=3→[1, 1], STOP=4→[0, 2], WAIT=5→[0, 1]. The numeric table has shape 6×2. GO STOP with two padding slots looks up[[2, 0], [0, 2], [0, 0], [0, 0]], mask[1, 1, 0, 0], shape 4×2. Its masked vector mean is[1, 1]; the all-slot mean is[.5, .5]. A real UNK position stays included. Squared distance sums matching-coordinate squared differences.

  For query GO[2, 0], exclude itself and reserved entries:TURN distance 2, STOP distance 8, WAIT distance 5, so TURN is nearest. For arbitrary query[1, 0], all four regular candidates are eligible:distances GO=1, TURN=1, STOP=5, WAIT=2; return both GO and TURN. For query STOP, distances to GO/TURN/WAIT are 8/2/1. The original coordinates were hand-chosen.

  The isolated GO-row objective toward[0, 2]has gradient[4, -4]; rate.25 gives[1, 1]and loss 8→2. Unchanged TURN also has[1, 1], so the updated rows collide without establishing useful meaning. GO STOP and STOP GO have the same mean[1, 1], also equal to the single TURN vector. Save/change the vocabulary mapping and table rows consistently. The pictured P/Q/R table uses separately supplied coordinates.

- For query [1, 0], compute squared distances to GO, TURN, STOP and WAIT in the original table. Return all nearest candidates.

> **My reasoning and evidence**

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Original mapping:PAD=0→[0, 0], UNK=1→[-1, -1], GO=2→[2, 0], TURN=3→[1, 1], STOP=4→[0, 2], WAIT=5→[0, 1]. The numeric table has shape 6×2. GO STOP with two padding slots looks up[[2, 0], [0, 2], [0, 0], [0, 0]], mask[1, 1, 0, 0], shape 4×2. Its masked vector mean is[1, 1]; the all-slot mean is[.5, .5]. A real UNK position stays included. Squared distance sums matching-coordinate squared differences.

  For query GO[2, 0], exclude itself and reserved entries:TURN distance 2, STOP distance 8, WAIT distance 5, so TURN is nearest. For arbitrary query[1, 0], all four regular candidates are eligible:distances GO=1, TURN=1, STOP=5, WAIT=2; return both GO and TURN. For query STOP, distances to GO/TURN/WAIT are 8/2/1. The original coordinates were hand-chosen.

  The isolated GO-row objective toward[0, 2]has gradient[4, -4]; rate.25 gives[1, 1]and loss 8→2. Unchanged TURN also has[1, 1], so the updated rows collide without establishing useful meaning. GO STOP and STOP GO have the same mean[1, 1], also equal to the single TURN vector. Save/change the vocabulary mapping and table rows consistently. The pictured P/Q/R table uses separately supplied coordinates.

- For the isolated GO-row update, give the gradient, updated vector, starting loss and new loss. Which other token now shares the vector?

> **My reasoning and evidence**

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Original mapping:PAD=0→[0, 0], UNK=1→[-1, -1], GO=2→[2, 0], TURN=3→[1, 1], STOP=4→[0, 2], WAIT=5→[0, 1]. The numeric table has shape 6×2. GO STOP with two padding slots looks up[[2, 0], [0, 2], [0, 0], [0, 0]], mask[1, 1, 0, 0], shape 4×2. Its masked vector mean is[1, 1]; the all-slot mean is[.5, .5]. A real UNK position stays included. Squared distance sums matching-coordinate squared differences.

  For query GO[2, 0], exclude itself and reserved entries:TURN distance 2, STOP distance 8, WAIT distance 5, so TURN is nearest. For arbitrary query[1, 0], all four regular candidates are eligible:distances GO=1, TURN=1, STOP=5, WAIT=2; return both GO and TURN. For query STOP, distances to GO/TURN/WAIT are 8/2/1. The original coordinates were hand-chosen.

  The isolated GO-row objective toward[0, 2]has gradient[4, -4]; rate.25 gives[1, 1]and loss 8→2. Unchanged TURN also has[1, 1], so the updated rows collide without establishing useful meaning. GO STOP and STOP GO have the same mean[1, 1], also equal to the single TURN vector. Save/change the vocabulary mapping and table rows consistently. The pictured P/Q/R table uses separately supplied coordinates.

- A developer swaps the IDs of GO and STOP but leaves the embedding rows unchanged. What goes wrong, and what must be saved or changed together?

> **My reasoning and evidence**

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A separate table contains A [0, 0], B [2, 0] and C [1, 2]. For query [1, 0], give all squared distances and every nearest candidate. Then calculate the mean of A and B and compare it with the query.

  Can that mean identify which ordered pair produced it?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Find a neighbourhood independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled embeddings and **neighbourhood**s **evidence** sheet to Image-pattern Network Investigation. Show how you can look up a vector using a token id and preserve sequence shape.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Calculate distances and a tie for Embeddings and **neighbourhood**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Embeddings and **neighbourhood**s. Proposed explanation: "Token ID differences define embedding distances". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A separate table contains A [0, 0], B [2, 0] and C [1, 2]. For query [1, 0], give all squared distances and every nearest candidate. Then calculate the mean of A and B and compare it with the query.

  Can that mean identify which ordered pair produced it?

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Measure embedding neighbours without turning numerical proximity into factual certainty.
- Toy table P =[0, 1], Q=[3, 1], R=[1, 4]; query [1, 1]. Use **squared Euclidean distance**.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

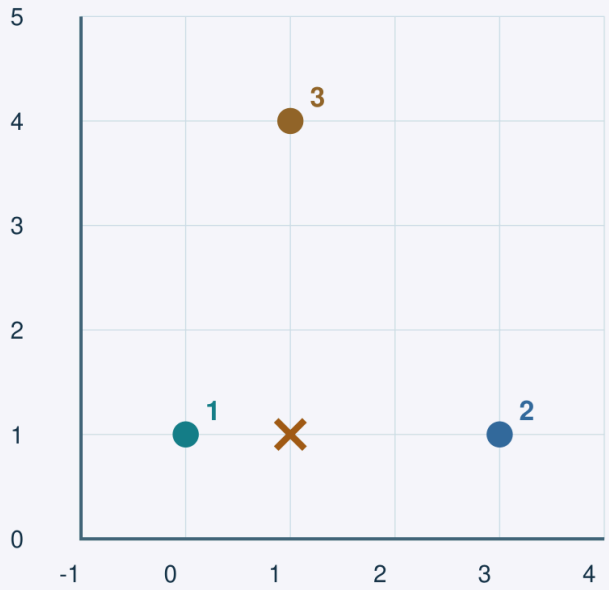
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/26

Squared Euclidean embedding distances



Horizontal: first embedding coordinate; vertical: second embedding coordinate

1: P at (0, 1)

2: Q at (3, 1)

3: R at (1, 4)

Cross: query at (1, 1)

Distance rule: Squared Euclidean distance

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/26

Embeddings and neighbourhoods

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Distance to P

(1-0)2+(1-1)2=1.

↓

2. Distances to Q/R

Q distance 4; R distance 9.

↓

3. Select neighbour

P is nearest under the declared metric; this is a geometric result, not proof of meaning.

CHANGE ONE THING

Move the query to [2, 1].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Measure embedding neighbours without turning numerical proximity into factual certainty.
- Which labelled step matches this detail? P is nearest under the declared metric; this is a geometric result, not proof of meaning.
- 1. Distance to P
- 2. Select neighbour
- 3. Distances to Q/R
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Select neighbour P is nearest under the declared metric; this is a geometric result, not proof of meaning. An embedding stores each item as an ordered vector.

Under **squared Euclidean distance**, subtract matching coordinates, square their differences and add. The smallest total identifies the nearest supplied vector.

These invented coordinates let us check geometry exactly; a claim that neighbours share meaning needs **evidence** about how the representation was learned and used.

Changed-case question and optional hint

- Changed-case practice: Move the query to [2, 1]. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Q becomes nearest with squared distance 1; P distance 4; R distance 10.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Q becomes nearest with squared distance 1; P distance 4; R distance 10. Changing the query position changes geometric neighbours while the stored table remains fixed.

Repair and retain the record

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower **training** loss alone establish that four updates generalise better?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Image tensors and channels (ai-5/20). Goal: Read a pixel using declared row column and channel axes. Short practice: Where is the bottom green pixel in the first batched image?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Embeddings and neighbourhoods: the case

AI ACADEMY / AI-5/26 · 01 · LOOK AT THE FACTS

Embeddings and neighbourhoods: the case

The situation

Measure embedding neighbours without turning numerical proximity into factual certainty.

Supplied evidence

Toy table P =[0, 1], Q=[3, 1], R=[1, 4]; query [1, 1]. Use squared Euclidean distance.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain embeddings and neighbourhoods using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

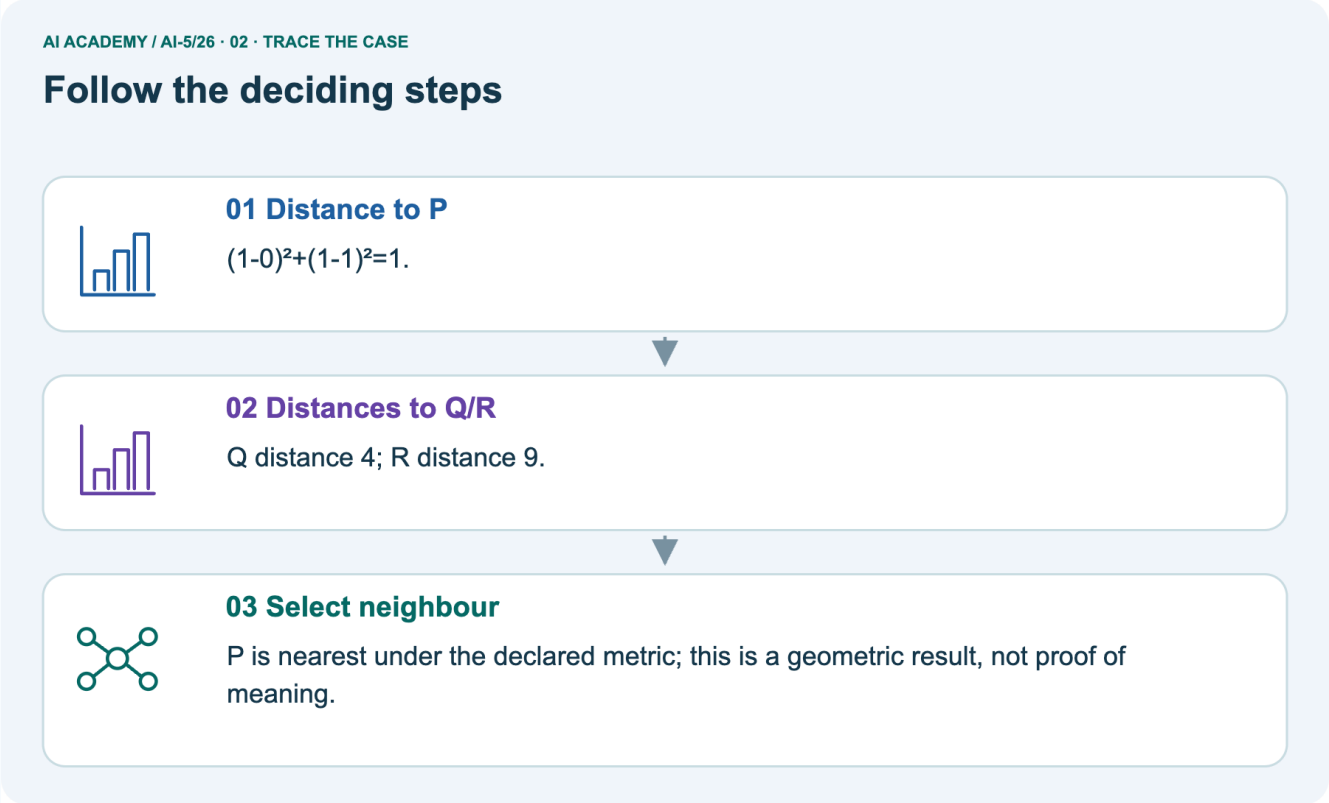
Measure embedding neighbours without turning numerical proximity into factual certainty.
Supplied facts: Toy table P =[0, 1], Q=[3, 1], R=[1, 4]; query [1, 1]. Use **squared Euclidean distance**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Distance to P: (1-0)²+(1-1)²=1. → Distances to Q/R: Q distance 4; R distance 9. → Select neighbour: P is nearest under the declared metric; this is a geometric result, not proof of meaning.

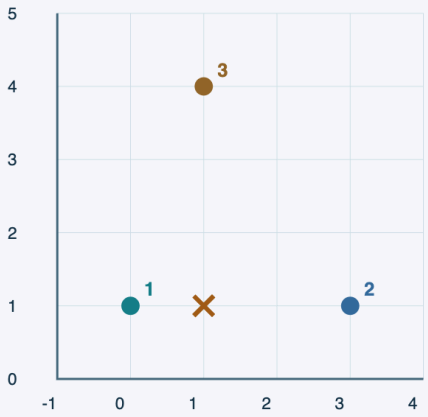
Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Squared Euclidean embedding distances

AI ACADEMY / AI-5/26

Squared Euclidean embedding distances



Horizontal: first embedding coordinate; vertical: second embedding coordinate

1: P at (0, 1)

2: Q at (3, 1)

3: R at (1, 4)

Cross: query at (1, 1)

Distance rule: Squared Euclidean distance

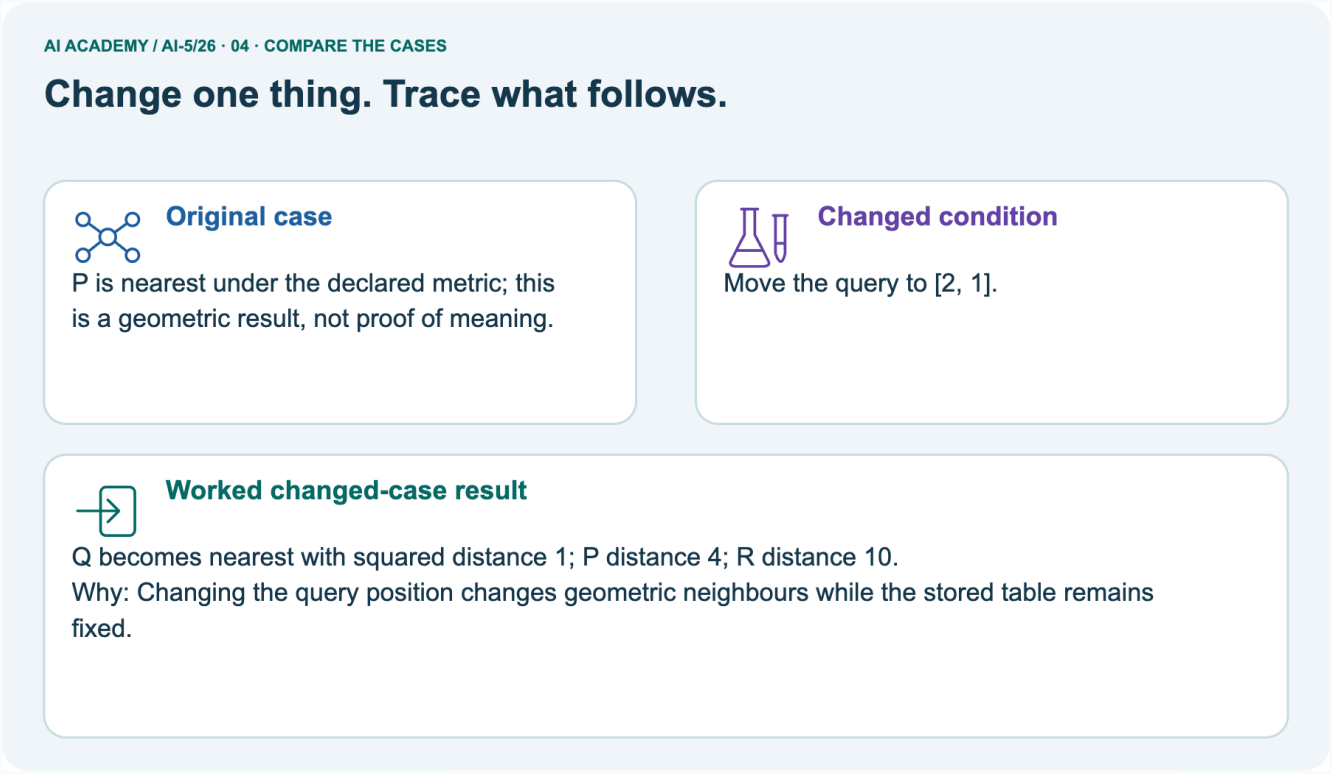
Follow the labelled picture with the matching explanation. Open the vector diagram

Squared Euclidean embedding distances. P: (0, 1). Q: (3, 1). R: (1, 4). Query query: (1, 1). Axes: first embedding coordinate; second embedding coordinate. Distance: **Squared Euclidean distance**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

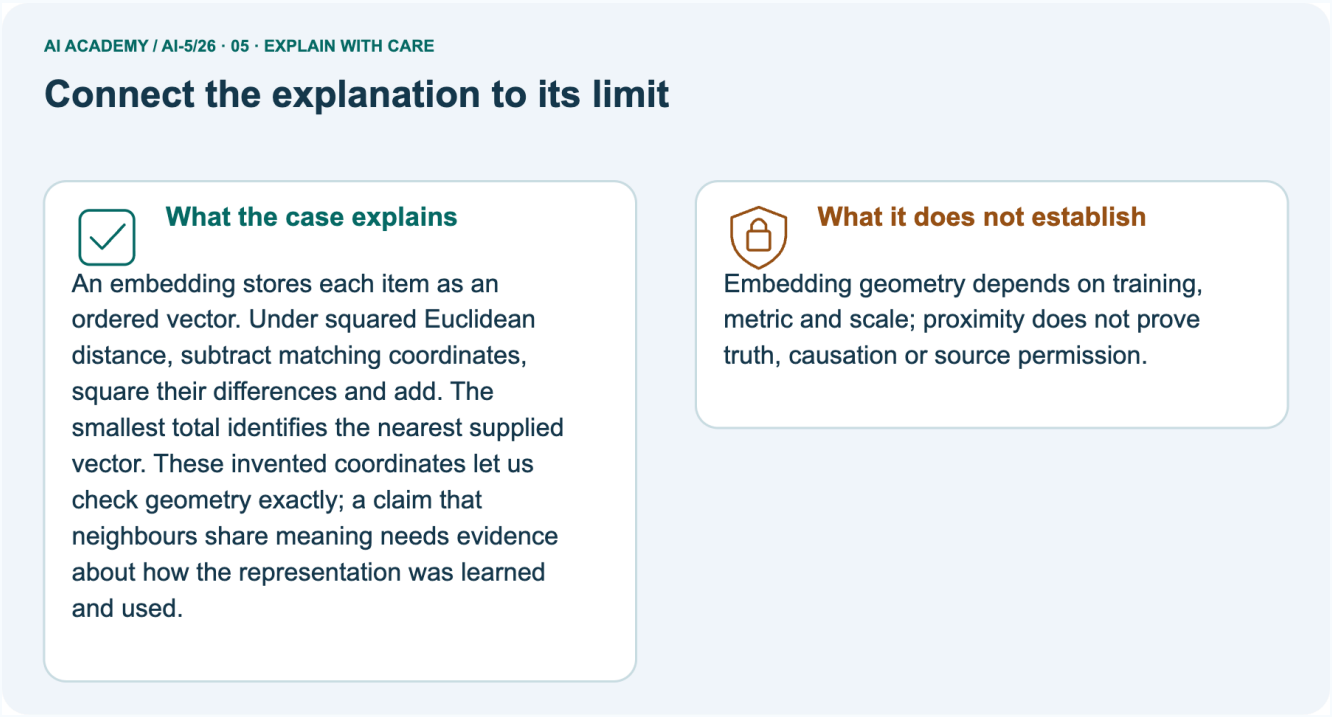
Original case: P is nearest under the declared metric; this is a geometric result, not proof of meaning. Changed condition: Move the query to [2, 1].

Expected result: Q becomes nearest with squared distance 1; P distance 4; R distance 10. Reason: Changing the query position changes geometric neighbours while the stored table remains fixed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An embedding stores each item as an ordered vector. Under **squared Euclidean distance**, subtract matching coordinates, square their differences and add. The smallest total identifies the nearest supplied vector.

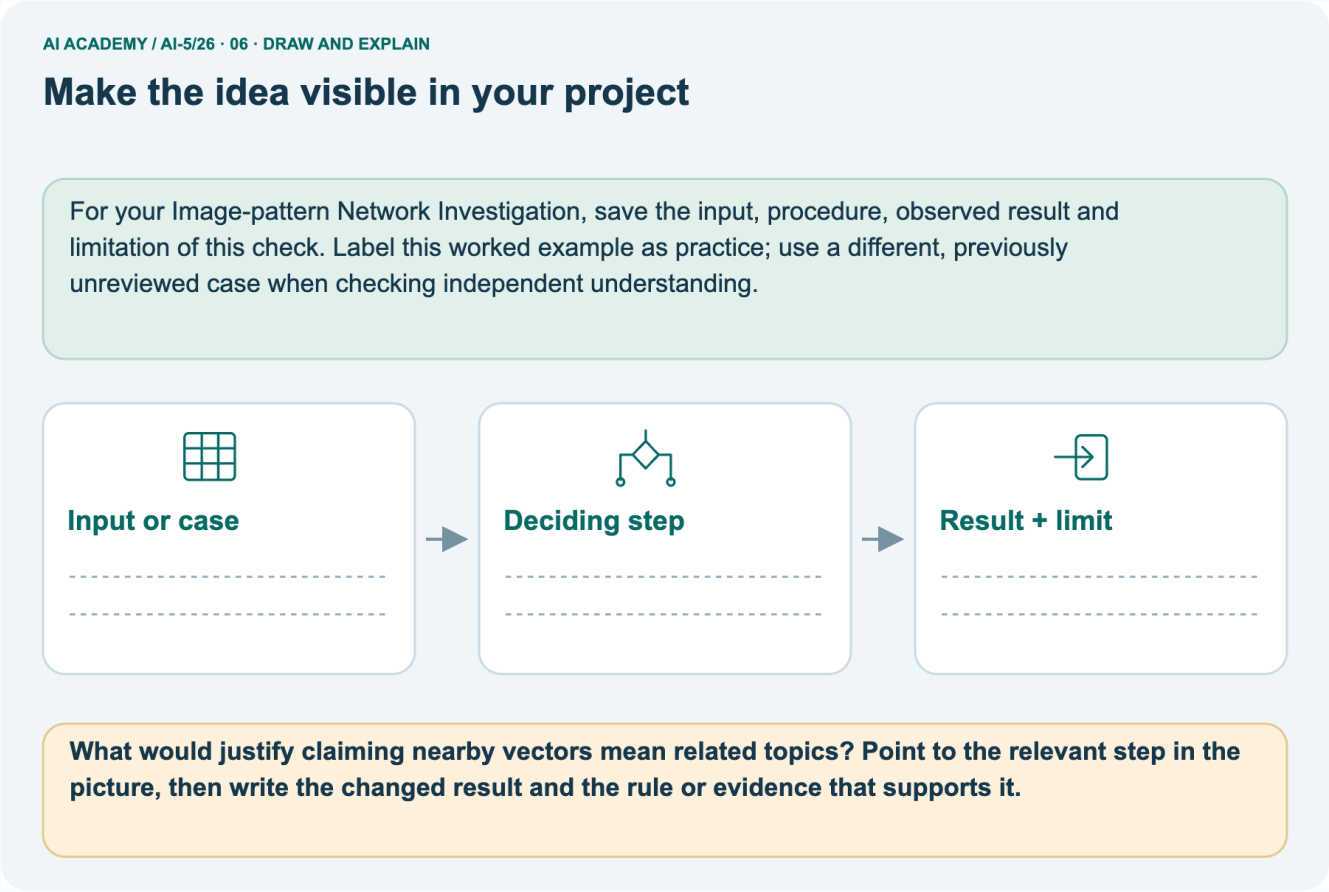
These invented coordinates let us check geometry exactly; a claim that neighbours share meaning needs **evidence** about how the representation was learned and used.

Limit: Embedding geometry depends on **training**, metric and scale; proximity does not prove truth, causation or source permission.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Embeddings and neighbourhoods: the case

02. Follow the deciding steps

03. Squared Euclidean embedding distances

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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